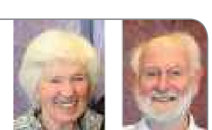
A common misconception about evolution is that individual organisms evolve. It is true that natural selection acts on individuals: Each organism's traits affect its survival and reproductive success compared with other individuals. But the impact of natural selection only becomes apparent in how a *population* of organisms changes over time.

To see why, consider the population of medium ground finches decimated by a drought, resulting in only 180 survivors out of some 1,200 birds. Researchers Peter and Rosemary Grant observed that during the drought, small, soft seeds were scarce, while large, hard seeds were more plentiful. Birds with larger, deeper beaks were better able to crack and eat the larger seeds and survived at a higher rate than finches with smaller beaks.

Since beak depth is an inherited trait in these birds, the offspring of surviving birds also tended to have deep beaks. As a result, the average beak depth in the next generation of *G. fortis* was greater than it had been in the pre-drought population **(Figure 23.2)**. The finch population had evolved by natural selection. However, the *individual* finches did not evolve. Each bird had a beak of a particular size, which did not grow larger during the drought. Rather, the proportion of large beaks in the population increased from generation to generation: The population evolved, not its individual members.

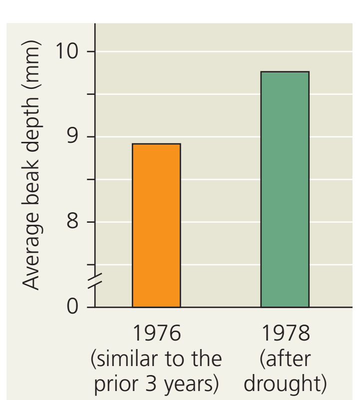
Focusing on evolutionary change in populations, we can define evolution on its smallest scale, called **microevolution**, as a change in allele frequencies in a population over generations. In the finch population, for example, birds that had alleles encoding large beaks survived at higher rates than did other birds—causing those alleles to be more common after the drought than they had been before it. There are three main mechanisms that can cause such changes in allele frequencies: natural selection, genetic drift (chance events that alter allele frequencies), and gene flow (the transfer of alleles between populations). Each of these mechanisms affects the genetic composition of populations, but only natural selection consistently improves the degree to which organisms are well suited for life in their environment (adaptation). Before we examine this more closely, let's revisit a prerequisite for these processes: genetic variation.

Mastering Biology
Interview with Rosemary and Peter Grant: Studying evolution in the Galápagos finches



▶ **Figure 23.2 Evidence of selection by food source.** The data represent adult beak depths of finches hatched before and after the 1977 drought. In one generation, natural selection resulted in a larger average beak size in the population.

Instructors: A related Experimental Inquiry Tutorial can be assigned in **Mastering Biology**.



CONCEPT 23.1

Genetic variation makes evolution possible

In *The Origin of Species*, Darwin provided abundant evidence that life on Earth has evolved over time, and he proposed natural selection as the primary mechanism for that change. He observed that individuals differ in their inherited traits and that selection acts on such differences, leading to evolutionary change. Although Darwin realized that variation in heritable traits is a prerequisite for evolution, he did not know precisely how organisms pass heritable traits to their offspring.

Just a few years after Darwin published *The Origin of Species*, Gregor Mendel wrote a groundbreaking paper on inheritance in pea plants proposing a model of inheritance in which organisms transmit discrete heritable units (now called genes) to their offspring. Although Darwin did not know about genes, Mendel's paper set the stage for understanding the genetic differences on which evolution is based.

Genetic Variation

Individuals within all species vary in their phenotypic traits. Humans, for example, vary noticeably in facial features, height, and voice. And though you cannot identify blood group (A, B, AB, or O) from a person's appearance, this and many other molecular traits also vary extensively among individuals.

Such phenotypic variations often reflect **genetic variation**, differences among individuals in the composition of their genes or other DNA sequences. Some heritable phenotypic differences occur on an "either-or" basis, such as the flower colors of Mendel's pea plants: Each plant had flowers that were either purple or white (see Figure 14.3). Characters that vary in this way are typically determined by a single gene locus, with different alleles producing distinct phenotypes. In contrast, other phenotypic differences vary in gradations along a continuum. Such variation usually results from the influence of two or more genes on a single phenotypic character. In fact, many phenotypic characters are influenced by multiple genes, including coat color in horses **(Figure 23.3)**, seed number in maize (corn), and height in humans.

How much do genes and other DNA sequences vary from one individual to another? Genetic variation at the

▼ **Figure 23.3 Phenotypic variation in horses.**